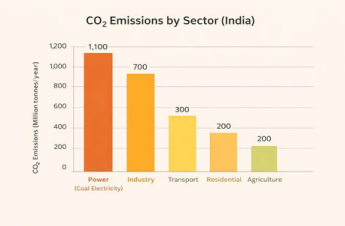
01. Public Health Risk due to Pollution

Pollution exposure trends show increasing mortality risk due to CO_2, SO_2, and urban emissions due to urbanisation.

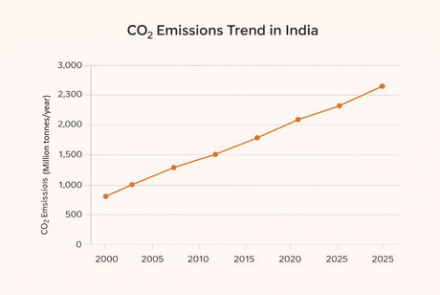
INSIGHTS

Chart Header



This bar chart displays sector-wise CO_2 emissions in India, highlighting that the power generation sector — primarily coal-based electricity — is the dominant contributor to national carbon emissions. Industrial activities form the second-largest emission source, followed by transport, residential energy use, and agriculture.

Chart Header



The chart highlights a consistent upward trajectory in CO_2 emissions, indicating increasing fossil fuel dependence despite renewable energy expansion. The sharpest growth period is observed between 2010 and 2020, reflecting accelerated industrial output and transport sector expansion.

POLLUTION RISK SCORE

48

Trends indicate long-term health impacts including respiratory disease burden and mortality risks.

WHY IS THIS

- Aging pipelines
- Poor sewage seperation
- Monitoring Failure

WHAT TO DO

- Smart IoT sensors
- Pipeline mapping
- Early outbreak warning system

IF POLLUTION TREND CONTINUES

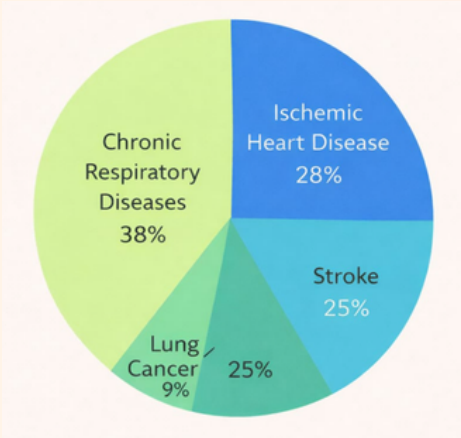
Healthcare System Overload:

If pollution exposure continues to rise, hospitals may face increased admissions due to respiratory and cardiovascular diseases, reducing emergency response capacity and increasing preventable mortality.

Chronic Disease Burden Escalation

Long-term exposure to polluted air and contaminated water can significantly increase chronic illness prevalence, leading to reduced workforce productivity and sustained economic pressure on public health infrastructure.

Chart Header

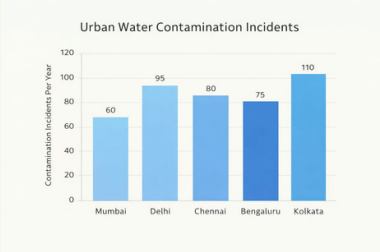


01. Recurring Uban Water Contamination Risk

Repeated contamination incidents across cities indicate systemic infrastructure vulnerability and growing public health exposure risks.

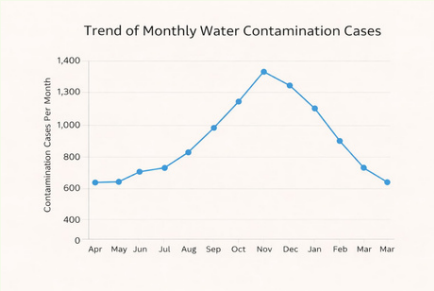
INSIGHTS

Chart Header



The chart indicates that densely populated urban centres such as Kolkata and Delhi experience higher contamination incident rates, suggesting increased vulnerability due to aging pipeline networks, high population density, and complex water distribution systems.

Chart Header



The chart demonstrates a clear upward trend in contamination incidents during the monsoon and post-monsoon months, with peak exposure observed between September and November. This seasonal surge reflects increased pipeline leakage, sewage overflow, and flood-related contamination risks in densely populated urban regions.

WATER RISK SCORE

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Moderate contamination exposure risk.

WHY IS THIS

- Infrastructure Updates
- Better Monitoring
- Policy Enforcement

WHAT TO DO

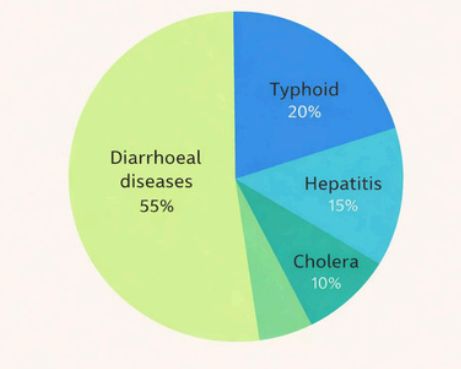
- Scale Sensors
- Expand Treatmenr
- Strengthen audits

FUTURE PREPAREDNESS OUTLOOK

Infrastructure Readiness:

Strengthened monitoring and upgraded distribution systems enable proactive contamination prevention and rapid response capability.

Sustained Risk Control:

Continuous investment in sanitation, surveillance, and digital monitoring ensures long-term stability in urban water safety.

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